
IoT Condition-Based Monitoring (CBM)

Training Guide for Residential & Commercial Building Systems

Version: 1.0

Prepared For: Operations, Service, Installation & Sales Personnel

Prepared By: _____

Date: _____

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Course Overview

Course Description

This course introduces the principles of Internet of Things (IoT) technology as it applies to Condition-Based Monitoring (CBM) of residential and commercial building systems. Participants will learn how connected sensors, cloud software, and analytics work together to improve equipment reliability, reduce maintenance costs, and support predictive maintenance strategies.

Intended Audience

- Service Technicians
 - HVAC Technicians
 - Electricians
 - Maintenance Personnel
 - Facility Managers
 - Building Engineers
 - Sales Engineers
 - Operations Managers
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Course Duration

Level	Duration
Basic	1 Day
Intermediate	2 Days
Advanced Certification	3 Days

Learning Objectives

Upon completion of this course, participants will be able to:

- Explain Condition-Based Monitoring concepts
- Describe IoT system architecture
- Select appropriate sensors for monitoring applications

- Install and commission IoT monitoring devices
- Configure gateways and cloud connectivity
- Build dashboards and alarms
- Interpret equipment health data
- Apply cybersecurity best practices
- Troubleshoot communication and hardware issues
- Develop monitoring strategies for residential and commercial facilities

Module 1 – Introduction to Condition-Based Monitoring

Learning Objectives

Participants will:

- Define CBM
- Compare reactive, preventive, predictive, and condition-based maintenance
- Understand return on investment (ROI)

Topics

- Equipment Reliability
- Asset Life Cycle
- Maintenance Strategies
- Downtime Costs
- Predictive Maintenance
- Remote Monitoring
- Return on Investment

Practical Exercise

Compare maintenance costs using:

- Reactive Maintenance
- Preventive Maintenance
- Condition-Based Maintenance

Module 2 – IoT Fundamentals

Topics

IoT Components

- Sensors
- Controllers
- Edge Devices
- Gateways
- Cloud Platform Mooreview
- Mobile Applications

Communication Technologies

- Wi-Fi
- Ethernet
- Bluetooth
- HaLoW
- Cellular (NB-IOT-CAT 1/LTE-CAT 4/5G)

Activity

Create an IoT architecture diagram for a commercial building.

Module 3 – Sensors Used in Condition Monitoring

Temperature

Applications

- HVAC
- Boilers
- Motors
- Bearings
- Refrigeration

Vibration/ MCSA

Applications

- Pumps
- Motors
- Fans
- Compressors

Current Monitoring

Applications

- Motor Load
- Runtime Monitoring
- Overcurrent Detection
- High current/extended start time on single phase motors

Voltage Monitoring

Applications

- Brownout Detection
 - Phase Loss
 - Power Quality
-

Pressure

Applications

- Air Compressors
 - Water Systems
 - Refrigeration
-

Flow

Applications

- Water
 - Gas
 - Air
-

Environmental

- Temperature
 - Humidity
 - CO₂
 - VOC
 - PM2.5
 - Carbon Monoxide
-

Leak Detection

- Water-In line
 - Water-Out of line-ropes-3 point
 - Refrigerant
-

Occupancy

- PIR
- Ultrasonic
- Radar

Practical Exercise

Match each sensor to the correct monitored asset.

Module 4 – Residential Applications

Topics include:

- Heat Pumps
- Air Conditioners
- Furnaces
- Water Heaters
- Sump Pumps
- Electrical Panels
- Indoor Air Quality
- Leak Detection
- Solar Systems
- Battery Storage
- Smart Home Integration

Case Study

Whole-home monitoring implementation.

Module 5 – Commercial Applications

Topics include:

- Commercial HVAC
- Chillers
- Cooling Towers
- Boilers
- Air Compressors
- Pumps
- Refrigeration Systems
- Lighting Systems
- Electrical Distribution
- Building Automation Systems (BAS)
- Backup Generators

Case Study

Multi-site office building monitoring.

Module 6 – Installation & Commissioning

Topics

- Site Surveys
- Sensor Placement
- Mounting Methods
- Wiring Practices
- Wireless Signal Testing
- Gateway Installation
- Calibration
- Functional Testing
- Documentation

Lab Exercise

Install sensors on demonstration equipment.

Module 7 – Connectivity & Networking

Topics

- IP Addressing
- DHCP
- Static IP
- Wi-Fi Configuration
- Ethernet Networks
- Cellular Connectivity
- VPN Access
- Firewalls
- Cloud Communications

Module 8 – Cloud Platforms & Dashboards

Topics

- Device Registration
- Dashboard Design
- Trend Charts
- Asset Management
- User Permissions
- Reports
- Mobile Applications
- API Integration

Exercise

Create a dashboard for : HVAC equipment.

Module 9 – Alarm Management

Topics

- Alarm Thresholds
- Warning vs Critical
- Escalation Procedures
- Notifications
- Alarm Suppression
- Workflows
- Documentation

Lab

Configure alarm rules for:

- High Temperature
- High Vibration
- Water Leak
- Power Failure

Module 10 – Data Analysis & Predictive Maintenance

Topics

- Baseline Measurements
- Trend Analysis
- Equipment Health
- Predictive Analytics
- Remaining Useful Life
- Energy Consumption
- Maintenance Recommendations

Exercise

Analyze six months of equipment trend data.

Module 11 – Cybersecurity

Topics

- Strong Passwords
- Multi-Factor Authentication
- Encryption
- Secure Remote Access
- Firmware Updates
- Device Authentication
- Network Segmentation
- Data Privacy

Module 12 – Troubleshooting

Topics

- Offline Devices
- Sensor Failures
- Gateway Issues
- Network Problems
- Incorrect Readings
- Calibration Errors
- Cloud Connectivity
- Firmware Recovery

Module 13 – Hands-On Capstone Project

Working in teams, students will:

- Conduct a site survey
- Select sensors
- Install equipment
- Configure the gateway
- Connect cloud software
- Build dashboards
- Configure alarms
- Analyze collected data
- Present recommendations

Assessment

Assessment	Weight
Module Quizzes	20%
Installation Lab	20%
Dashboard Configuration	15%
Alarm Configuration	15%
Troubleshooting Exercise	10%
Final Capstone Project	20%

Equipment Required

- IoT Gateway
- Wireless Sensors
- Current Transformers (CTs)
- Temperature Sensors
- Pressure Sensors
- Vibration Sensors
- Water Leak Sensors
- Laptop
- Tablet
- Mobile Commissioning App (MV web)
- Network Tester

- Digital Multimeter
- Basic Hand Tools
- PPE

Appendix A – Common IoT Devices

Environmental Monitoring

- Temperature Sensors
- Humidity Sensors
- Air Quality Sensors
- CO₂ Sensors
- VOC Sensors

Mechanical Monitoring

- Vibration Sensors
- Bearing Temperature Sensors
- Pressure Transmitters
- Flow Meters

Electrical Monitoring

- Energy Meters
- Current Transformers
- Voltage Sensors
- Power Quality Monitors

Infrastructure

- IoT Gateways
- Edge Controllers
- Cellular Routers
- Smart Relays

Appendix B – Glossary

- **CBM** – Condition-Based Monitoring
 - **IoT** – Internet of Things
 - **BAS** – Building Automation System
 - **CMMS** – Computerized Maintenance Management System
 - **CT** – Current Transformer
 - **VPN** – Virtual Private Network
 - **MFA** – Multi-Factor Authentication
 - **RUL** – Remaining Useful Life
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